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A.2 Conditional Probability

1 Conditional Probability

Eg. $P(\text{roll 3 or less} \mid \text{odd}) = \frac{(\text{3 or less}) \text{ and } (\text{odd})}{\text{odd}}$

$= \frac{\{1,3\}}{\{1,3,5\}} = \frac{2}{3}$

Eg.

	HD	No HD	
atleast 1 parent	95	180	275
no parent	65	295	360
	160	475	

Total = 635

$P(\text{No HD \& no parent}) = \frac{295}{635}$

Eg. $P(\text{No HD} \mid \text{no parent}) = \frac{295}{360}$

- Conditional Probab. of A given B is,

$$\boxed{P(A \mid B) = \frac{P(A \cap B)}{P(B)}}$$

- Also,

$P(B) \times P(A \mid B) = P(A \cap B) = P(A) \times P(B \mid A)$

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- The key words that help you to identify, it is a conditional probab. question.
  → given sth. is this, it is possible to have the other thing by 80%.
  → 80% clg students use FB
  $$P(FB \mid Clg) = 0.8$$
- When a lang. restricts the possible outcomes to one grp./case it means that the conditioning is present on that case.

Eg.

0.08	0.42	0.29	0.79	Reg.
0.08	0.06	0.07	0.21	Irreg.
0.16	0.48	0.36		
L	N	H		

$$P(\text{Reg and } L\ BP) = 0.08$$

2. <u>Independence</u>

- Earlier, we saw ⇒ $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$
  $$P(A \cap B) = P(B) \times P(A \mid B) = P(A) \times P(B \mid A)$$
- Now, A and B are said to independent if,
  $$\boxed{P(A \cap B) = P(A) \times P(B)}$$

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:- That also means, that
$P(A) = P(A|B)$ or $P(B) = P(B|A)$, thus,
knowing if A or B has occured gives no information on whether or not the other event has occured.

Eg. Indep. event → A & B. $P(A) = 0.6$
$P(A \cap B) = 0.3$

$$P(B) = \frac{P(A \cap B)}{P(A)} = \frac{0.3}{0.6} = 0.5$$

$$P(A|B) = P(A) = 0.6$$

Eg. $P(A) = 0.4$, $P(B) = 0.1$, $P(A \cap B) = 0.05$

Are the independent? → No

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{0.05}{0.4} = \frac{5 \times \cancel{10}}{\cancel{100}_{2} \times 4}$$

$$= 0.125$$

Eg. $P(A) = 0.2$, $P(B) = 0.3$

(i) indep. $P(A \cup B) = 0.2 + 0.3 - 0.06$
$= \cancel{0.56}\ 0.44$

(ii) mutually exclusive → $P(A \cap B) = 0$

$$P(A \cup B) = 0.5$$

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3. Sequence of Events

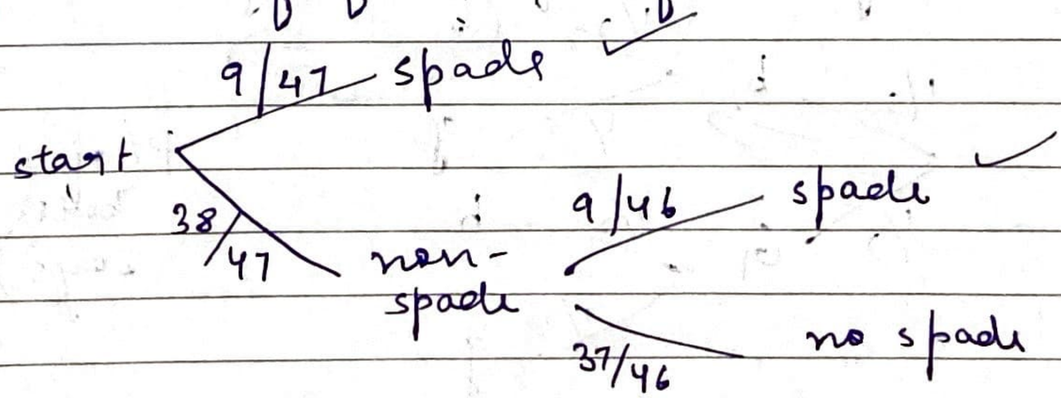
- $P(B|A) = \frac{P(B \cap A)}{P(A)} \Rightarrow P(A) \times P(B|A) = P(A \cap B)$

it occurs as a seq. of events, first A should occur & then B need to occur, taking into account how A has occurred.

Eg. find the probab. of having a flush after dealing with 5 cards from a std. deck.

$$P(\text{all 5 cards with same suit}) = 4 \times P(\text{all are from suit})$$
$$= 4 \times \frac{13}{52} \times \frac{12}{51} \times \frac{11}{50} \times \frac{10}{49} \times \frac{9}{48} = 0.198\%$$

Eg. if exactly 4 of first 5 cards are spades, probab. of flush in first 7 cards.



$$\frac{9}{47} + \frac{38}{47} \times \frac{9}{46} = 0.35$$

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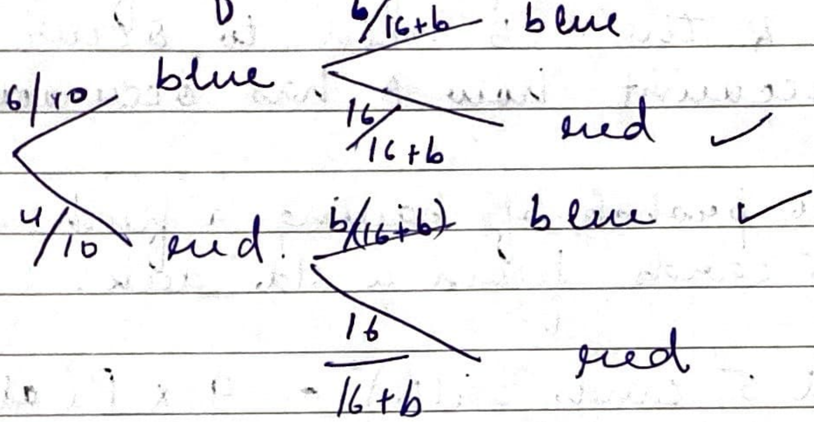
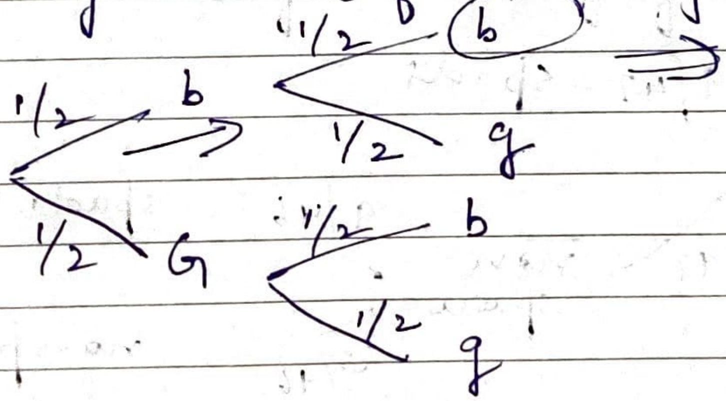
Eg. Urn has 10 balls: 4 red & 6 blue.
2 Urn has 16 red balls & unknown blue balls.
Prob. that both balls (drawn each from diff. urns) are diff. colour is 0.528
no. of blue balls in urn 2.



$$0.528 = \frac{6}{10} \times \frac{16}{(16+b)} + \frac{4}{10} \times \frac{b}{(16+b)}$$

solve for b $b = 9$.

Eg. 2 children, not twins. Given at least 1 is boy. Prob. (of both boys)



$P(\text{of both boys}) = \frac{1}{2} \times \frac{1}{2}$

$= \frac{1}{4}$

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4. <u>Bayes' Theorem</u>

- we often want $P(A|B)$ but are given $P(B|A)$.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\boxed{P(A|B) = \frac{P(A) \times P(B|A)}{P(B)}}$$

→ Bayes' Theorem

↳ this can be calculated by law of total probability. By adding up the P(B) by breaking into cases.

Eg.

Age	P(A|Age)	Assumed
16-20	0.06	0.08
21-30	0.03	0.15
31-65	0.02	0.49
66-99	0.04	0.28

$$P(\underset{31-65}{Age}|A) = \frac{P(Age) \times P(A|Age)}{P(A)}$$

$$= \frac{0.02 \times 0.49}{\Sigma P(A \text{ of all age grps})}$$

$$= 0.3234$$

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- Law of Total Probab.

if $A_1 \dots A_k$ are disjoint & $\sum_{i=1}^{k} P(A_i) = 1$ then;

$$P(B) = P(B \cap A_1) + \dots + P(B \cap A_k)$$
$$P(B) = P(A_1) \times P(B|A_1) + \dots + P(A_k) \times P(B|A_k)$$

These sets $A_1 \dots A_k$ are known as the partitions of sample space. This is also referred as the list of all possible cases.

Eg. 50% S, 40% P, & 10% UP

$P(D|S) = 0.01$, $P(D|P) = 0.005$, $P(D|UP) = 0.001$

$$P(S|D) = \frac{P(S) \times P(D|S)}{P(D)}$$
$$= \frac{0.5 \times 0.01}{0.5 \times 0.01 + 0.4 \times 0.005 + 0.1 \times 0.001}$$
$$= \frac{0.005}{0.005 + 0.002 + 0.0001}$$
$$= \frac{0.005}{0.0071} \doteq 70\%$$